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• SOLAR ENERGY • LESSON 09

Solar Wiring Basics

Safe, Correct Wiring from Panel to Load

Lesson 09 of 30 | Solar Energy Programme | AYE Tech Hub

Awet G. Nway

Founder, AYE Tech Hub | Solar & Electrical Engineering Expert

- Complete solar system wiring overview: panels → MPPT → battery → inverter → loads
- Series vs parallel panel wiring — voltage, current and when to use each
- DC cable sizing: voltage drop calculation, IEC/NEC standards, fuse selection
- MPPT charge controller wiring: connections, settings and safety rules
- Battery bank wiring: series, parallel and series-parallel configurations
- AC output wiring: inverter to distribution board, breakers and earthing
- Grounding, earthing and lightning protection for PV systems

Solar PV Wiring — Overview and Safety First

Wiring is where most solar installation mistakes happen — and where most solar fires and electrical hazards originate. Correct solar wiring requires understanding DC electrical circuits, cable sizing, overcurrent protection, and grounding. **DC electricity at solar system voltages (24–96V) and currents (5–50A) can cause severe burns, fires, and electrocution.** Unlike AC, DC arcs do not self-extinguish at zero-crossing — a DC arc started by a loose connection or undersized fuse can sustain itself and ignite a fire. Correct wiring technique is not optional — it is life safety.

Complete Solar PV Wiring Flow — DC Side and AC Side

Fig 1 — Solar system wiring: all DC components on left, inverter converts to AC for household loads

The Four Wiring Sections of a Solar System

Section	From → To	Voltage Type	Key Hazards	Protection Required
Panel to MPPT (Array DC)	Solar panel junction box → MPPT charge controller PV input	DC — variable (Voc to Vmpp)	High open-circuit voltage; reverse polarity burns MPPT	PV fuse/combiner; surge protection device (SPD)
MPPT to Battery (Battery DC)	MPPT output → Battery positive and negative terminals	DC — regulated (battery voltage)	High current; thermal runaway risk from wrong cable size	DC circuit breaker; correct cable cross-section
Battery to Inverter	Battery bank → Inverter DC input terminals	DC — battery voltage (12/24/48V)	Very high short-circuit current (200–2000A); arc flash	Fused disconnect as close to battery as possible
Inverter to AC Distribution Board	Inverter AC output → Main distribution board (DB)	AC — 230V/50Hz	Electric shock; overload of inverter; backfeed hazard	MCB at DB; RCD; proper earthing of neutral and PE

SAFETY RULE — Always install a DC isolator switch between the solar array and the MPPT, and another between the battery and inverter. These allow safe isolation of each section for maintenance without risk. Never work on the array side while it is exposed to sunlight — cover panels with an opaque tarpaulin first.

Series vs Parallel Panel Wiring

Before wiring panels to the MPPT charge controller, you must decide how to connect them — in series, in parallel, or in a series-parallel combination. This decision is driven by the MPPT's input voltage range and current rating, the system voltage (12V, 24V, or 48V), and the total array power. Getting this wrong either damages the MPPT or leaves power on the table.

Panel Wiring Configurations: Series vs Parallel

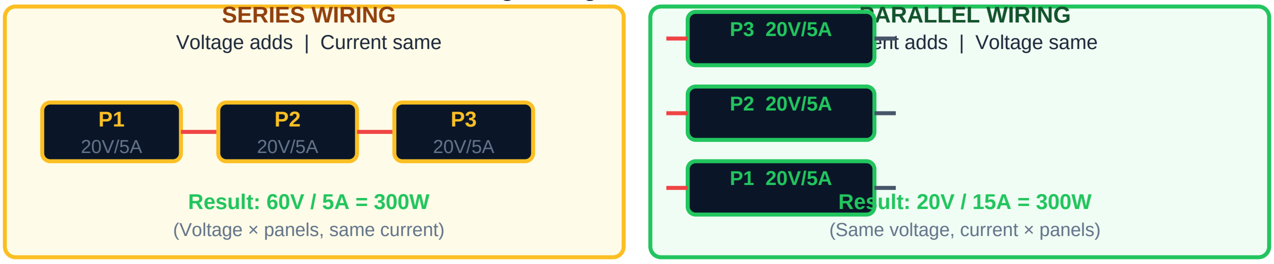

Fig 2 — Series wiring increases voltage; parallel wiring increases current. Choose based on inverter/MPPT input voltage requirement.

Connection	Effect on Voltage	Effect on Current	Use When	Risk if Wrong
Series	Multiplies: V_total = V × N panels	Same as one panel: I_total = I_panel	MPPT requires higher input voltage (48V+ systems). Reduces cable losses at same power	If total Voc exceeds MPPT max input voltage → MPPT destroyed instantly
Parallel	Same as one panel: V_total = V_panel	Multiplies: I_total = I × N panels	MPPT is limited to low voltage (12/24V battery bank) but handles high current	Each string needs its own fuse — without fuses, panel strings can backfeed each other → fire
Series-Parallel	Intermediate (balanced V and I)	Balanced between series and parallel	Larger arrays where neither pure series nor parallel works within MPPT limits	Must balance strings — mismatched strings reduce output and cause reverse current

Critical MPPT Voltage Rules

MPPT Parameter	What It Means	Consequence of Violation
Maximum PV Input Voltage (e.g., 150V)	Highest voltage the MPPT can safely receive from the array	Exceeding this destroys the MPPT immediately and permanently — no repair possible
MPPT Voltage Range (e.g., 30–120V)	Voltage range in which the MPPT can track maximum power	If array voltage outside this range, MPPT either does not charge or charges at reduced efficiency

Maximum PV Input Current (e.g., 20A)	Maximum DC current from the array the MPPT can handle	Exceeding this blows internal fuses or damages MPPT input stage
Open Circuit Voltage (Voc)	Maximum panel voltage at zero current (cold, full sun) — NOT operating voltage	Must calculate total string Voc at minimum temperature — Voc rises as panels get colder

TEMPERATURE WARNING — Panel Voc increases as temperature decreases. If you are in a highland area (Tigray evenings can reach 5–10°C in winter), calculate Voc at the lowest expected temperature: Voc_cold = Voc_STC × [1 + Temperature_Coefficient × (T_min – 25°C)]. For a typical panel with Voc=38V and temp coefficient –0.3%/°C at 5°C: Voc_cold = 38 × [1 + (–0.003) × (5–25)] = 38 × 1.06 = 40.3V per panel. Calculate this before finalising the number of panels in series.

DC Cable Sizing — The Most Critical Calculation

Undersized DC cables are the number one cause of solar system fires. DC cable must carry the full system current continuously (unlike AC cables that have zero-crossing relief). Cable sizing must account for three things: **current-carrying capacity** (ampacity), **voltage drop**, and **short-circuit protection**. A cable that passes the ampacity test may still fail the voltage drop limit — always check both.

Step-by-Step DC Cable Sizing

#	Step — Formula / Rule	Example (48V / 1kW system)
1	Find system current: I = P / V_system, then add 25% safety margin	I = 1000/48 = 20.8A → design for 26A
2	Check IEC ampacity table: select cross-section where rated current > design current	4mm² rated 32A (free air) → PASS
3	Calculate voltage drop: ΔV = (2 × L × I × ρ) / A, ρ_Cu = 0.0172 Ω·mm²/m	L=10m, I=26A, A=4mm²: ΔV = 2.24V
4	Check drop limit: ΔV% = (ΔV / V_system) × 100% — limit 1–3% for DC	2.24/48 × 100 = 4.7% → FAIL, too high
5	Upsize cable until ΔV% ≤ 3%	6mm²: ΔV=1.49V → 3.1% → borderline
6	Final selection — use next size up for margin and future expansion	10mm²: ΔV=0.89V → 1.86% → PASS
7	Select fuse: rating = 1.25 × I_SC (panel short-circuit current per string)	I_SC=5.5A × 2 strings × 1.25 = 13.75A → use 15A DC fuse

Cable Cross-Section	Max Current (single, free air)	Max Current (bundled/conduit)	Typical Solar Application
1.5 mm²	17A	13A	Small panel strings (≤10A), signal wires
2.5 mm²	23A	18A	Panel strings up to 20A, 12V small systems
4 mm²	32A	24A	Panel to MPPT runs ≤15m, 24V systems
6 mm²	40A	31A	Panel to MPPT standard run, 48V systems

10 mm²	55A	42A	Battery to inverter ≤1000W, long runs
16 mm²	73A	56A	Battery to inverter 1000–2000W systems
25 mm²	95A	73A	Battery to inverter 2000–3000W, main bus
35 mm²	119A	89A	Battery to inverter 3000–5000W, commercial
50 mm²	145A	110A	Large inverter connections, low-voltage high-current systems

USE SOLAR-RATED CABLE ONLY — Solar DC cables (EN 50618 / TUV 2Pfg 1169) are rated for outdoor UV exposure, high temperatures (90–120°C), and DC voltage. Standard AC house wire is not suitable for solar strings — it will degrade within 2–3 years under UV and heat, causing insulation failure and fire risk.

MPPT Charge Controller Wiring — Correct Connection Order

The order in which you connect the MPPT charge controller matters critically. Connecting the solar array before the battery causes a voltage spike that can destroy the MPPT's internal circuits instantly. Always follow the sequence below regardless of what brand of MPPT you are using:

Step	Connection to Make	Why This Order
1st — BATTERY	Battery (+) and (–) → MPPT battery terminals	MPPT reads battery voltage first — initialises internal circuits safely before any other source
2nd — LOAD	Load output terminals → connected loads	Loads connect with battery already powering MPPT — avoids floating output voltage spikes
3rd — SOLAR PV	Panel array (+) and (–) → MPPT PV input	PV connects last — MPPT is fully initialised and ready to accept array voltage without damage
DISCONNECT (reverse)	PV first → loads → battery last	Battery stays connected until PV is isolated — provides voltage reference for safe shutdown

MPPT Settings After Connection

Setting	Typical Value	How to Configure	Why It Matters
Battery Type	Sealed / AGM / Flooded / Lithium (LiFePO4)	Select from MPPT menu or DIP switches	Determines charging voltage profile — wrong setting overcharges or undercharges battery
Battery Voltage	12V / 24V / 48V	Auto-detected or manual selection	MPPT must know system voltage to calculate state of charge and set correct charge voltages
Absorption Voltage	LiFePO4: 3.65V/cell × cells (14.6V for 4S)	Set per battery manufacturer datasheet	Maximum charge voltage — exceeding this damages or causes thermal runaway in lithium batteries

Float Voltage	LiFePO4: 3.35V/cell (13.4V for 4S 12V bank)	Set per manufacturer	Maintenance voltage after full charge — prevents overcharge while panel is still producing
Load Control Mode	Manual / Low Voltage Disconnect (LVD)	Set LVD to 11.5V (12V system) or 23V (24V)	Prevents deep discharge — discharging below LVD damages lead-acid batteries permanently
Equalisation	Off for lithium; 1×/month for flooded lead-acid	Enable/disable in menu	Equalisation at high voltage destroys lithium batteries — must be disabled for LiFePO4

Battery Bank Wiring — Series, Parallel and Series-Parallel

Multiple batteries can be connected in series to increase voltage, in parallel to increase capacity (Ah), or in series-parallel to achieve both. **Rule: always use identical batteries** — same brand, model, age, and state of charge — when connecting multiple units. Mixing batteries creates imbalanced charging, premature ageing, and in lithium systems, dangerous cell imbalance.

Configuration	Battery Example	Result	Use For	Key Requirement
Single battery	1 × 12V/100Ah LiFePO4	12V / 100Ah / 1.2kWh	12V systems, small loads only	N/A
2 × Series	2 × 12V/100Ah in series	24V / 100Ah / 2.4kWh	24V system voltage	Same battery — connect (+) of one to (–) of next
4 × Series	4 × 12V/100Ah in series	48V / 100Ah / 4.8kWh	48V system (recommended for 3kW+)	Use a BMS rated for 48V (16S LiFePO4) or 4-cell connection
2 × Parallel	2 × 24V/100Ah in parallel	24V / 200Ah / 4.8kWh	Increasing runtime at same voltage	Equal cable lengths from each battery to bus bar — prevents current imbalance
2S × 2P	4 × 12V/100Ah: 2 series × 2 parallel	24V / 200Ah / 4.8kWh	Medium systems needing both voltage and capacity	Connect series strings first, then connect strings in parallel at the bus bar

FUSING RULE FOR BATTERY BANK — Install an ANL fuse or DC circuit breaker rated at 125% of maximum inverter DC input current, positioned within 30 cm of the battery positive terminal. The battery can deliver thousands of amps into a short circuit — a fuse this close to the battery is the only thing that prevents a catastrophic arc flash or battery fire.

Battery Cable Sizing — Based on Inverter Power

Inverter Power	Battery Voltage	Max DC Current	Min Cable Size	Fuse Rating

500W	12V	50A (with losses)	10mm²	60A ANL
1000W	12V	100A	16mm²	125A ANL
1000W	24V	50A	10mm²	60A ANL
2000W	24V	100A	16mm²	125A ANL
3000W	48V	75A	10mm²	100A ANL
5000W	48V	125A	25mm²	150A ANL
8000W	48V	200A	35mm²	250A ANL

Grounding, Earthing and AC Output Wiring

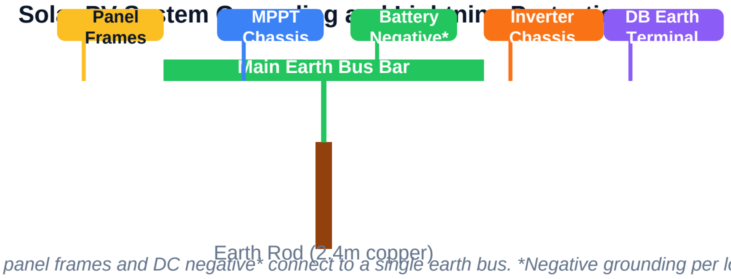

Fig 3 — All metal chassis, panel frames and DC negative connect to a single earth bus. *Negative grounding per local regulations.*

Why Grounding Is Critical in Solar Systems

A solar PV system without proper grounding is dangerous in three ways: (1) Panel frame faults can make the entire array frame live at lethal voltage; (2) Without a reference ground, induced lightning energy has nowhere to go safely except through your equipment and potentially your body; (3) Ungrounded systems in areas with frequent lightning (including highland Ethiopia) suffer very high equipment failure rates from transient overvoltages.

Grounding Connection	Cable Size	Standard	Purpose
Panel frame to earth bus	6mm² green/yellow	IEC 62548	Fault protection — prevents frame becoming live if panel insulation fails
MPPT chassis to earth bus	6mm² green/yellow	IEC 62548	Equipment protection — reference point for surge protection
Battery negative* to earth bus	10mm² green/yellow	IEC 62548	System reference earth — establishes 0V potential reference for DC system
Inverter chassis to earth bus	6mm² green/yellow	IEC 60364	AC safety earth — required for connection to AC distribution
AC neutral (N) to earth at DB	As per AC cable	IEC 60364-5-54	TN-S system — single neutral-earth bonding point at the distribution board only
Earth rod to main earth bus	16mm² bare copper	IEC 62561	Dissipates fault current and lightning energy safely into the earth

Lightning arrestor to earth bus	As short as possible — max 0.5m	IEC 61643	Shunts lightning transient to earth before it reaches equipment

AC Output Wiring — Inverter to Distribution Board

Inverter Rating	AC Output Cable	MCB at DB	RCD Required?	Notes
≤1,000W / 4.5A	2.5mm² 3-core (L/N/E)	10A	Yes — 30mA	Single socket circuit
1,000–2,300W / 10A	2.5mm² 3-core	16A	Yes — 30mA	Standard household circuit
2,300–3,500W / 16A	4mm² 3-core	20A	Yes — 30mA	Heavy loads — heaters, pumps
3,500–5,000W / 22A	6mm² 3-core	32A	Yes — 30mA	Sub-distribution board recommended
5,000–8,000W / 35A	10mm² 3-core	40A	Yes — 30mA	Commercial-grade distribution
Three-phase 10kW+	4mm² per phase + earth	3-phase MCCB	Yes — 100mA	Requires 3-phase inverter and balanced loads

Common Solar Wiring Mistakes — and How to Avoid Them

Mistake	Risk Level	Consequence	Correct Practice
Connecting PV array before battery to MPPT	CRITICAL	Instant MPPT destruction (internal voltage spike)	Always connect battery first, then load, then PV — strict order
Reverse polarity on any connection	CRITICAL	Immediate component destruction; possible fire	Test polarity with multimeter before every connection. Use red for +, black for –
Undersized DC cables	CRITICAL	Cable overheating, insulation melt, fire	Calculate ampacity AND voltage drop. Size for 125% of operating current
No fuse between battery and inverter	HIGH	Battery short circuit → uncontrolled arc → fire	ANL fuse within 30cm of battery positive — non-negotiable
Mixing old and new batteries in parallel	HIGH	Older battery drains newer; accelerated ageing of both; possible overcharge	Always use identical batteries: same brand, capacity, age, and SOC when connecting
No DC isolator between array and MPPT	HIGH	Cannot safely disconnect array for maintenance — live at high voltage	Install rated DC isolator on array positive. Cover panels before opening isolator
Loose MC4 connectors	HIGH	High resistance at connection → localised heating → fire	Use correct MC4 crimping tool. Click until you hear and feel positive lock. Test pull-force

No surge protection device (SPD)	MODERATE	Lightning transient destroys MPPT, inverter, and connected equipment	Install Type 2 DC SPD rated for system Voc at MPPT input terminals

KEY TAKEAWAYS — SOLAR-009: Solar Wiring Basics 1. Wire in sections: Array → MPPT → Battery → Inverter → AC Distribution. 2. Series wiring raises voltage; parallel wiring raises current. Match to MPPT specification. 3. Size DC cables for both ampacity AND voltage drop (limit ≤3%). 4. Connect MPPT in order: battery first, load second, PV last — always. 5. Fuse within 30cm of battery positive terminal — non-negotiable. 6. Ground all metal chassis and panel frames to a single earth bus connected to a rod. 7. Next lesson — SOLAR-010: Solar Load Calculation — sizing the system for real loads.